QBRAID AWARD · 1 OF 3

What qBraid did for us

We built an eavesdropper alarm for a power-grid control link, using the BB84 quantum key distribution protocol.

QBRAID LAB
Qiskit notebook
Wrote and debugged the BB84 circuit

→

QBRAID SIMULATOR
GPU state vector
Tuned the alarm line on 30 qubits

→

SUPERCONDUCTING QPU
IQM Garnet
2 jobs: clean link and eavesdropper

→

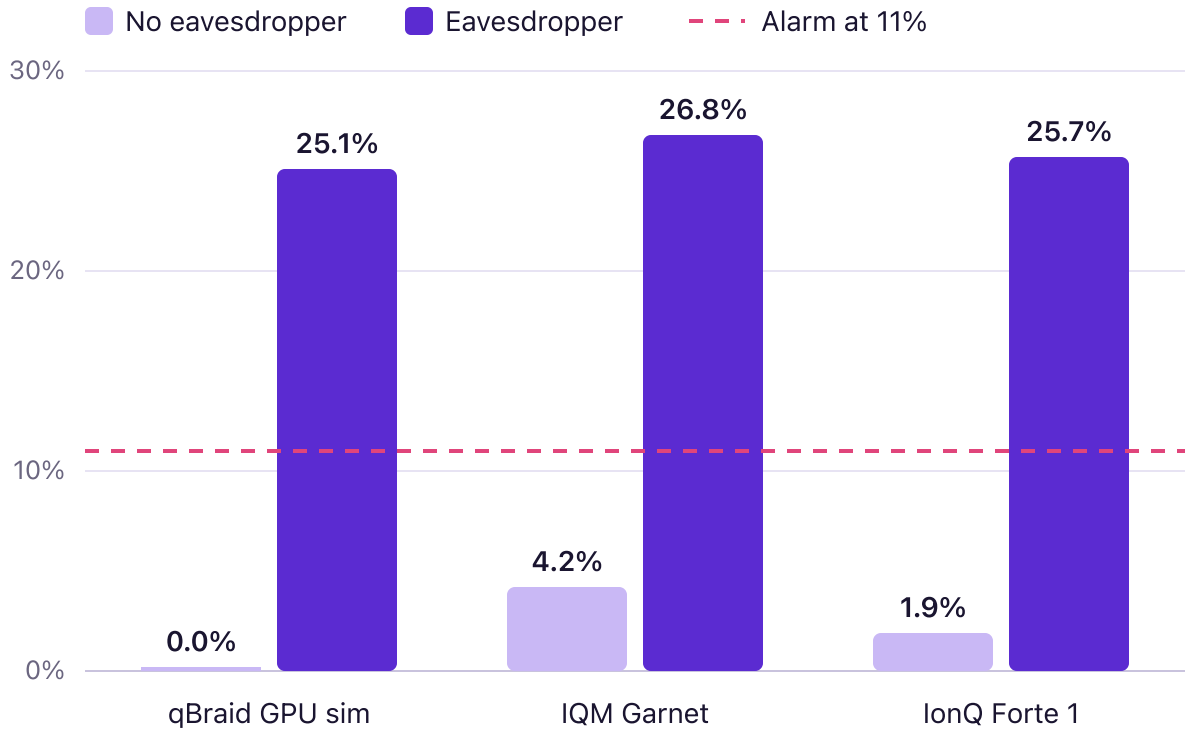
TRAPPED-ION QPU
IonQ Forte 1
2 jobs to cross-check a second kind of hardware

> **What qBraid made possible:** the same circuit on a simulator and two kinds of quantum hardware, from one notebook, with no rewrites.

We packed every key bit into one batched circuit per run, so each device needed 2 jobs instead of one per qubit.

QBRAID AWARD · 2 OF 3

Our best qBraid moment



Illustrative numbers. Error rate = share of key bits that disagree.

Noise alone stays under the alarm line. An eavesdropper pushes every backend past 25%.

So the alarm works on real quantum hardware, not just in simulation.

IQM Garnet's noise sits higher than IonQ's, but still leaves a wide margin below 11%.

Error rate: the share of key bits Alice and Bob disagree on after comparing bases.

QBRAID AWARD · 3 OF 3 · OPTIONAL

Check it

PROOF

<job ID 1> · IQM Garnet
<job ID 2> · IQM Garnet
<job ID 3> · IonQ Forte 1
<job ID 4> · IonQ Forte 1

Copied from the Jobs page in our qBraid account.

2-minute screen recording: <link>

RUN IT YOURSELF

Launch on qBraid

github.com/<team>/grid-guardians

CREDITS USED

GPU simulator, about 6 minutes	30
IQM Garnet, 2 jobs × 500 shots	205
IonQ Forte 1, 2 jobs × 50 shots	860
Total: about $11, 4% of our budget	**1,095**

Batching kept it cheap: 4 hardware jobs in total.

NEXT

Run the alarm on Quantinuum, and find how low the alarm line can go before noise alone sets it off.